

End Sem

Q1. Which of the following best describes a *confound* in experimental research design?

- A) A measurement that cannot be replicated across trials
 - B) A variable that the researcher intentionally manipulates
 - **C) An unmeasured variable that systematically varies with the IV and may affect the DV**
 - D) A random error that occurs during data collection
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Q2. A psychologist studies sleep deprivation on problem-solving. Participants in the 4-hour group report stress from staying in the lab overnight. Best way to control this confound?

- A) Increase the number of participants in the sleep-deprived group
 - B) Allow participants to choose their sleep condition
 - **C) Ensure both groups spend the night in the lab, regardless of sleep duration**
 - D) Use a different task to measure problem-solving ability
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Q3. A clinical trial gets $p=0.03$ (frequentist) but $BF_{10}=1.2$ (Bayesian). What does this difference imply?

- A) Both methods provide strong evidence for the alternative
 - B) p-value and Bayes Factor always agree when sample size is large
 - **C) p-value indicates statistical significance, while BF suggests only weak evidence for the alternative**
 - D) The Bayes Factor invalidates the frequentist result due to a mismatch in priors
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Q4. Logistic regression on exercise (Yes/No) with age and screen time. A **negative** coefficient for screen time means:

- **A) As screen time increases, the chance of exercising decreases, holding age constant**
- B) Screen time predicts lower actual exercise time in hours

- C) A negative coefficient means screen time has no effect on exercise likelihood
 - D) The model assumes screen time is unrelated to the outcome
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Q5. Linear regression predicts anxiety from screen time. $R^2=0.64$. Best interpretation?

- A) 64% of anxiety scores are predicted without any error
 - **B) 64% of the variation in anxiety scores is explained by screen time**
 - C) 64% of participants had high anxiety due to screen time
 - D) Screen time causes anxiety in 64% of people
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Q6. Regression: $\text{Score} = 30 + 4 \times (\text{Hours Studied}) + 2 \times (\text{Hours Slept})$. Best interpretation of the coefficient for *Hours Studied*?

- A) Students who study more always score higher than those who sleep more
 - B) Each additional hour studied increases the score by 4 points, regardless of sleep
 - **C) For each additional hour studied, the expected score increases by 4 points, holding hours slept constant**
 - D) For each additional hour studied, the expected score increases by 2 points, holding hours studied constant
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Q7. $\text{IQ} \sim \text{Normal}(\mu=100, \sigma=15)$. What % of participants have IQ between 85 and 115?

- A) 34%
 - **B) 68%**
 - C) 95%
 - D) Insufficient information
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Q8. Which is **NOT** a reason to use ANOVA instead of multiple t-tests?

- A) Control for Type I error
- B) Allows post-hoc comparisons
- **C) Higher chance of finding significance**
- D) Tests multiple means simultaneously

Q9. What is the purpose of Levene's Test in the context of ANOVA?

- A) To test for normality across the data
 - **B) To compare the variance across data conditions**
 - C) To analyze the effect size of the findings
 - D) To find outliers in the data
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Q10. Drug Y crossover trial: placebo → drug, then drug → placebo one year later. Most appropriate test?

- A) t-test
 - B) Repeated Measures ANOVA
 - **C) Mixed ANOVA**
 - D) Paired t-test
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Q11. Which best differentiates PCA from Factor Analysis for a 20-item psychological well-being questionnaire?

- A) PCA assumes items are caused by latent factors; FA extracts uncorrelated components
 - **B) PCA maximizes variance explained by components; FA models underlying latent constructs**
 - C) PCA is confirmatory; FA is exploratory
 - D) PCA requires larger sample size than FA
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Q12. Between-subjects study: satisfaction (ordinal: low/medium/high), two independent groups (n=25 each). Most appropriate nonparametric test?

- A) Wilcoxon Signed-Rank Test — compares paired ordinal data
 - **B) Mann-Whitney U Test — compares independent groups on an ordinal outcome**
 - C) Chi-Squared Test — analyzes continuous data between groups
 - D) Kruskal-Wallis Test — compares two independent groups
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Q13. Studying association between gender (male/female) and therapy preference (yes/no), $n=200$. Most appropriate condition for applying Chi-Squared Test of Independence?

- A) The data must be normally distributed
- B) The sample size must exceed 300
- C) The variables must be measured on an interval scale
- **D) None of the above**

6 descriptive questions from exam:

Question 1 (6 marks)

A university assigns 200 students to 4 study technique groups (Self-study, Coaching, Online courses, Group study). They run 6 separate t-tests at $\alpha=0.05$.

- a) What is the problem with running six separate t-tests without adjusting the significance level? (1 mark)
 - b) Explain what it means to control the family-wise error rate and how Bonferroni correction achieves it. (2 marks)
 - c) What do False Discovery Rate (FDR) correction methods control? (1 mark)
 - d) Mention one advantage and one disadvantage of each — Bonferroni and Benjamini-Hochberg corrections. (2 marks)
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Question 2 (7 marks)

Reaction times compared between caffeine vs placebo group. Data is heavily skewed with outliers (t-test assumptions violated). Use a permutation-based non-parametric test.

1. What is the null hypothesis? (1 mark)
 2. Key steps involved in performing the test. (5 marks)
 3. How is the p-value calculated from the permutation null distribution? (1 mark)
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Question 3 (8 marks)

Health psychologist data: stress score (0–40), caffeine intake (mg), physical activity group (Sedentary/Moderate/High), insomnia (Yes/No), sick days. For each sub-question, identify the most appropriate statistical method and justify.

- a) Does caffeine intake differ across the three physical activity groups? (2 marks)
 - b) Is there an association between insomnia and physical activity level? (2 marks)
 - c) Can stress scores be predicted from caffeine intake? (2 marks)
 - d) What is the best way to model sick days as a function of stress score and insomnia status? (2 marks)
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Question 4 (4 marks)

Give examples explaining when to choose a one-tailed vs two-tailed hypothesis test. Provide scenarios where one is more appropriate than the other.

Question 5 (4 marks)

What is multicollinearity and how does it impact interpretation of a multiple regression model? Given that PCA and FA both help address multicollinearity, what are the essential differences between them? Illustrate with an example.

Question 6 (3 marks)

Between-subjects study: stress levels (1–5 ordinal scale), mindfulness vs control group (n=25 each), data and residuals are not normally distributed. What statistical test would you suggest and why? List potential challenges.

Key:

1. C
2. C
3. C
4. A

5. B
6. C
7. B
8. C
9. B
10. C
11. B
12. B
13. D

Here are the solutions:

Question 1 (6 marks)

a) Problem with 6 separate t-tests (1 mark)

Each test has a 5% chance of a false positive. With 6 tests, the probability of getting *at least one* false positive is:

$$P(\text{at least one false reject}) = 1 - (0.95)^6 \approx 0.265$$

So you have a ~26.5% chance of a false positive — far above the intended 5%. This is the **multiple comparisons problem**.

b) FWER and Bonferroni (2 marks)

FWER = probability of making *even one* false positive across all comparisons. We want this ≤ 0.05 .

Bonferroni correction:

- With n tests, adjust the significance threshold to α/n
- Here: $0.05/6 \approx 0.0083$
- Only reject H_0 for a test if its p-value ≤ 0.0083
- Equivalently, multiply each p-value by n and compare to α

This ensures $P(\text{false reject}) = 1 - (1 - 0.05/n)^n \leq 0.05$.

c) FDR (1 mark)

FDR controls the **proportion of false discoveries among all rejected hypotheses** (all "significant" results).

$$\text{FDR} = \text{FP} / (\text{FP} + \text{TP}) = \text{false positives} / \text{total discoveries}$$

Unlike Bonferroni which controls making even one false positive, FDR allows some false positives but keeps their *rate* controlled. Less conservative → more statistical power.

d) Advantages & Disadvantages (2 marks)

	Bonferroni	Benjamini-Hochberg
Advantage	Strongly controls FWER — virtually no false positives	Less conservative — higher power, fewer false negatives
Disadvantage	Too strict — increases false negatives (misses real effects)	Doesn't control FWER strongly — can produce more false positives if p-values are correlated

Shared disadvantage: Both assume independent tests, which is often violated in practice.

Question 2 (7 marks)**1. Null Hypothesis (1 mark)**

Let μ_0 = observed difference in means (caffeine – placebo) on original data.

H₀: There is no difference in mean reaction times between the caffeine and placebo groups. ($\mu_{\text{caffeine}} - \mu_{\text{placebo}} = 0$)

2. Key Steps (5 marks)

- 1. Compute observed statistic:** Calculate $\mu_0 = \text{mean}(\text{caffeine}) - \text{mean}(\text{placebo})$ on original data.

2. Repeat for B = 10,000 iterations:

- Merge both groups into one pool
- Randomly split into two groups of the original sizes
- Compute μ = difference in means of the two new groups
- If $\mu_0 > \mu$, increment counter by 1

3. Compute p-value: $p = \text{count} / 10,000$

4. Decision: If $p \leq 0.05$, reject $H_0 \rightarrow$ significant difference. Otherwise, accept H_0 .

5. Visualization: Plot the bootstrap distribution of μ across all iterations. Check whether μ_0 lies outside the 95% confidence interval of this null distribution.

3. P-value from permutation null distribution (1 mark)

$p\text{-value} = (\text{number of permutations where permuted statistic} \geq \text{observed } \mu_0) / (\text{total number of permutations})$

It represents how likely the observed group difference is under the null. If $p < 0.05$, the observed difference is unlikely to have occurred by chance.

Question 3 (8 marks)

Assumption stated in answer: All tests are parametric (normality assumed, all data points independent).

a) Caffeine intake across 3 physical activity groups (2 marks)

- **IV:** Physical activity group $\rightarrow$ categorical (3 levels)
- **DV:** Caffeine intake $\rightarrow$ numerical continuous

Method: One-way ANOVA

Justification: Comparing means of a continuous DV across 3 independent categorical groups. ANOVA is appropriate for multiple (>2) group comparisons while controlling Type I error. Assumes homogeneity of variance (check with Levene's test).

b) Association between insomnia and physical activity level (2 marks)

- **IV:** Physical activity group → categorical (3 levels)
- **DV:** Insomnia → categorical (Yes/No), can be converted to count per group

Method: Chi-Squared Test of Independence

Justification: Both variables are categorical. Chi-squared tests whether the distribution of insomnia frequency is consistent across the 3 activity groups. Non-parametric — no distributional assumptions needed. Compares observed vs expected cell frequencies.

c) Predicting stress scores from caffeine intake (2 marks)

- **IV:** Caffeine intake → numerical continuous
- **DV:** Stress score → numerical (0–40), modelled as continuous

Method: Simple Linear Regression

Justification: Both predictor and outcome are numerical. Linear regression models the linear relationship between them. Assumes normality of residuals, homoscedasticity, and independence.

d) Modelling sick days from stress score + insomnia status (2 marks)

- **IV:** Stress score → numerical continuous; Insomnia status → categorical (can be 0/1)
- **DV:** Sick days → numerical count

First-pass answer: Multiple Linear Regression — since predictors and outcome are all numerical (insomnia converted to 0/1), and assuming normality of residuals.

Better answer: GLM with Poisson regression + log link

Justification: Sick days is a *count variable* — non-negative integers, residuals need not be normal. A GLM with Poisson family and log link handles count data properly without forcing normality assumptions.

Question 4 (4 marks)

One-tailed test: Tests for significance in one direction only (higher or lower).

Use when:

- You have strong prior belief or evidence that the effect is in one specific direction
- The test statistic distribution is asymmetric (e.g., F-distribution) — only the upper tail is meaningful

Example: "Does exam preparation *improve* scores?" — you only care if scores go up, not down. One-tailed is justified.

Two-tailed test: Tests for significance in both directions. Use when:

- No prior assumption about direction of effect
- Both tails of the distribution carry meaningful interpretation

Example: "Does a drug *affect* blood pressure?" — it could go up or down. Two-tailed is appropriate.

General rule: Ideally always use a two-tailed test unless there is strong, pre-specified directional justification. In regression/ANOVA, the overall F-test is one-sided by nature, but post-hoc comparisons are two-tailed.

Question 5 (4 marks)

Multicollinearity occurs when multiple predictors in a regression model are highly correlated with each other.

Impact on regression:

- Regression coefficients become unstable and hard to interpret — a single predictor's coefficient can't be isolated cleanly
- Standard errors inflate → coefficients may appear non-significant even when they are (FN increases)
- Model may overfit to noise in redundant dimensions
- AIC/model fit appears inflated

Detect using **VIF (Variance Inflation Factor)** — $VIF > 5$ indicates problematic multicollinearity.

PCA vs FA to address multicollinearity:

	Factor Analysis (FA)	PCA
Goal	Uncover hidden latent constructs that <i>cause</i> the observed variables	Project data onto new axes that <i>maximize explained variance</i>
Nature	Confirmatory/suggestive — researcher has theory about latent structure	Exploratory — no assumed latent structure
Factors/Components	Factors may be correlated, no fixed number	Components are orthogonal (uncorrelated), fixed
Example	Questionnaire: hidden variables like "anxiety" and "depression" drive responses to questions — FA finds these	Personality traits: many correlated scores → PCA finds the fewest components that explain most variance

Question 6 (3 marks)

Setting: Ordinal outcome (1–5), two independent groups, n=25 each, non-normal data and residuals.

Recommended test: Mann-Whitney U Test

Justification:

- Two independent groups → not paired
- Outcome is ordinal → ranks can be computed
- Data and residuals are non-normal → parametric t-test is inappropriate
- Mann-Whitney is the non-parametric alternative to the independent samples t-test; computes the U statistic based on ranks, determines if one group tends to have higher values than the other

Challenges:

- Small sample size (n=25) — borderline for even non-parametric tests; if the data were less skewed, a t-test might actually be robust enough here
- Confounds not accounted for — any unmeasured variables (e.g., baseline stress, medication) could bias results

- Ordinal scale reliability — the 1–5 scale assumes equal perceived intervals between levels, which may not hold
- The test tells you *if* groups differ in rank distribution, but not the magnitude of the effect (need additional effect size measure like rank biserial)